

Classical Machine Learning Cheat Sheet

Problem Types

Classification: categorical targets. Regression: continuous targets. Clustering: unlabeled data. Dimensionality Reduction: PCA/LDA. Anomaly Detection: Isolation Forest.

Model Selection

Linear/Logistic Regression for simple baselines; Decision Trees for interpretability; Random Forest and XGBoost/LightGBM/CatBoost for strong tabular performance; SVM for small datasets; KNN for distance-based tasks; Naive Bayes for text.

Scaling

Scale: Linear/Logistic Regression, SVM, KNN, PCA, K-Means, Neural Nets. Don't scale: Decision Trees, Random Forest, XGBoost, LightGBM, CatBoost.

Standardization vs Normalization

Standardization (mean=0, std=1) for optimization-based models. Normalization (0-1) for KNN, K-Means, image pixels, some neural nets.

Evaluation Metrics

Classification: Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC. Regression: MAE, MSE, RMSE, R^2 . Clustering: Silhouette Score.

Encoding

Binary→Label, Nominal→One-Hot, Ordinal→Ordinal Encoding, High-cardinality→Target/CatBoost Encoding.

Quick Rules

Use Recall for disease detection, Precision for spam/fraud, Cross-validation for model comparison, Tree models don't require scaling.